

Store Sales Data and Modeling Report

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1 Introduction

In this project, we will be working with the Store Sales Time Series data set from Kaggle. The data set can be found at <https://www.kaggle.com/competitions/store-sales-time-series-forecasting>. The goal of the competition is to accurately predict the sales for the last sixteen days of August 2017. The provided data for training contains all the sales prior to the test dates, dating back to the beginning of 2013. The competition uses root mean squared logarithmic error (RMSLE) as its evaluation metric.

2 EDA

Since the data has already been provided on Kaggle, there is minimal cleaning that must be done. The relevant provided files are train.csv, test.csv, holidays_events.csv, stores.csv, and oil.csv. Of all the files, only oil.csv contains missing values. About 30.88% of the oil values are missing for all the given dates. It is also seen in Figure 1 that the oil prices start off around 100 but at the end of 2014 drop to be around 50. From Figure 2, one sees that the sales, on average, seem mostly unaffected by this drop. For these reasons, and in order to keep our models more simple, we choose not to utilize the oil data.

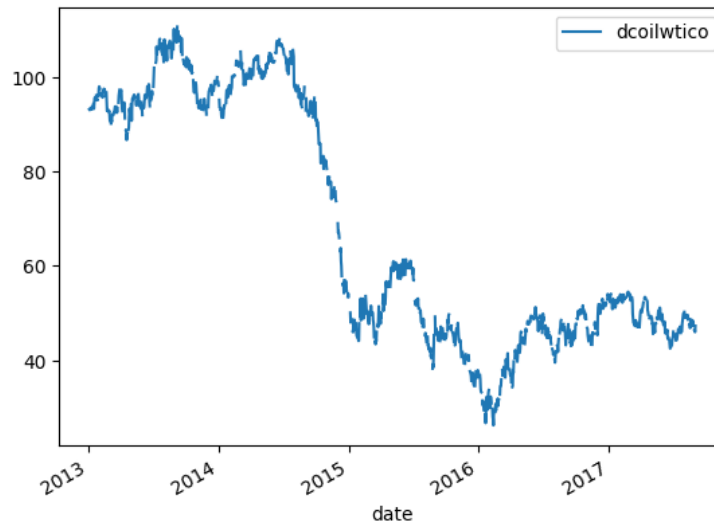


Figure 1: Oil Prices

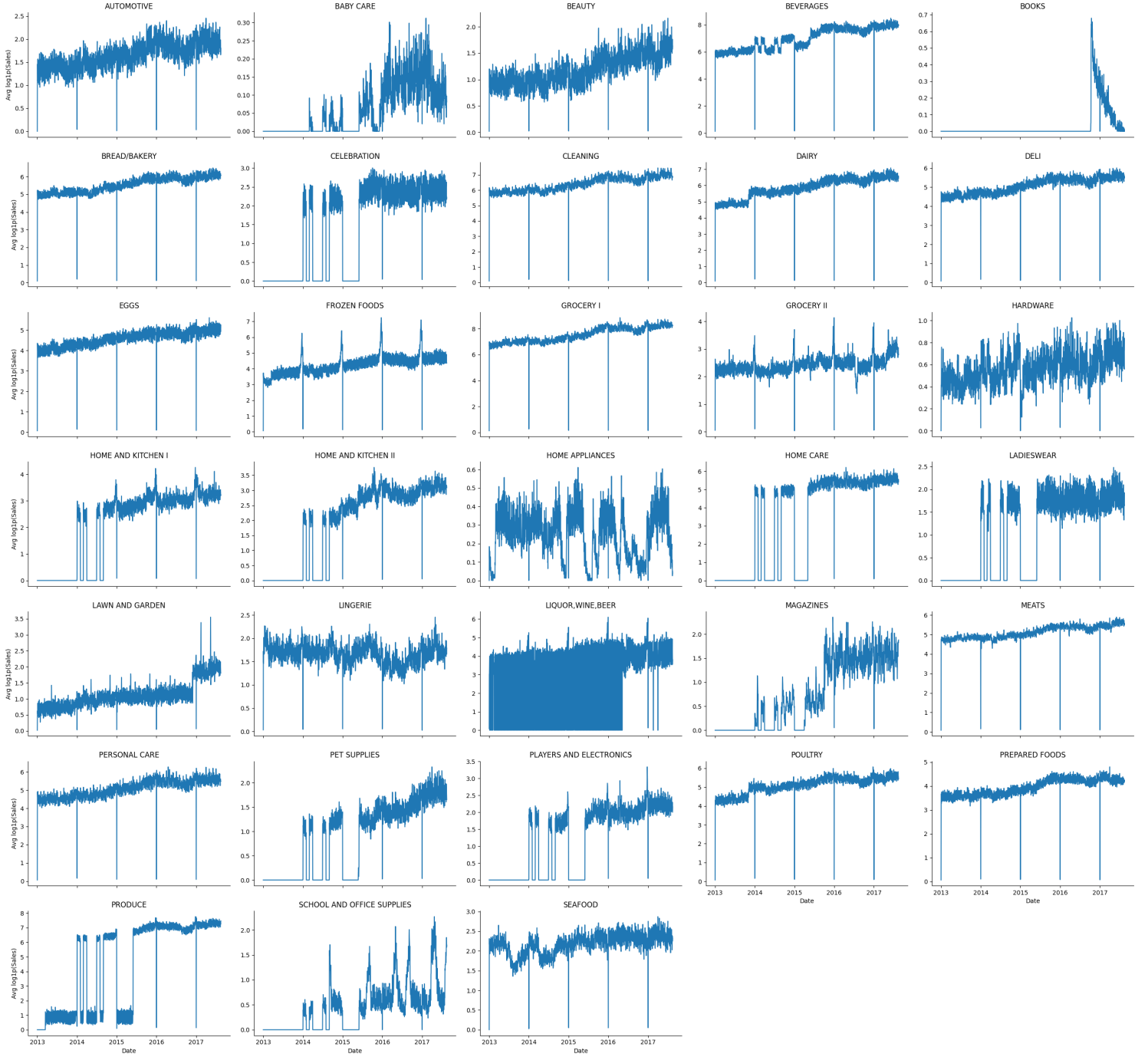


Figure 2: Average $\log_{1p}(\text{sales})$ by family and date

The last bit of analysis on the data that we do is examining the relationship between the store numbers, clusters, and types. Each store has a unique number that identifies it. There are 54 different stores. Each store then falls into one of 17 clusters and one of 5 types. The only cluster which contains stores of different types is cluster 10, which contains stores of type B, D, and E. Every other cluster only contains stores of the same type. Thus, we will view clusters as being more specific than types.

3 Feature Engineering

For all models, we log-transformed the target variable using $\log_{1p}(\text{sales})$. This approach provides two main advantages: (1) it reduces the skewness of the sales data, and (2) computing the RMSE on $\log_{1p}(\text{sales})$ perfectly aligns with the competition's evaluation metric of RMSLE. To obtain the final sales predictions, we simply applied the inverse transformation (`expm1`).

Additionally, all models utilized the product family and store number as categorical features.

Starting with Model 2, we introduced the day of the week as a categorical variable, alongside sine and cosine transformations of the day of the year (with periods of 1, 1/2, and 1/3 of a year) to capture seasonality. To account for external events, we included a boolean feature indicating if a date falls within a month of the April 16, 2016 earthquake. Because public sector wages are paid on the 15th and the last day of the month, we added a feature tracking the number of days since the last payday, as well as a calendar-based holiday indicator. Finally, we incorporated 16-, 21-, and 28-day lags of the target variable. Since the Kaggle test set is 16 days long, restricting our lags to 16+ days allows us to forecast the entire window directly without relying on recursive predictions. The 16-day lag represents the most recent available data, while the 21- and 28-day lags correspond to exactly three and four weeks prior, capturing weekly shopping habits.

For Model 4 and subsequent iterations, we included an interaction term between the product family and the number of items on promotion. This captures the reality that promotional sensitivity varies significantly by product category (e.g., a promotion on beverages might yield a different sales spike than one on books).

Prior to building Model 5, we utilized Lasso regression to perform feature selection on various lag intervals, which led us to include additional 17-, 18-, 19-, and 20-day lags, alongside interaction and polynomial terms for most numeric features. These specific additions were exclusive to Model 5.

For Models 6, 7, and 8, we experimented with different interaction terms to capture spatial and categorical nuances. Model 6 introduced an interaction between family type and store type; Model 7 used family type and store cluster; and Model 8 used family type and store number. These features were designed to account for the fact that specific product families sell differently depending on a store’s specific region, size, or customer demographic.

4 Modeling

To establish a baseline (Model 1), we predicted the historical average of $\log_{1p}(\text{sales})$ for each unique combination of store and product family. Model 2 expanded on this by fitting independent linear regressions for each store-family pair, while Model 3 applied Ridge regression to these same groups to introduce regularization. However, while Models 2 and 3 achieved low RMSLE scores on our internal validation sets, they performed poorly on the hidden test set. This discrepancy strongly suggested overfitting and an inability to capture macro-level trends. Consequently, starting with Model 4, we transitioned to a global approach: training a single Ridge regression model across the entire dataset. For all models, prior to final evaluation, we clipped our predictions to bound them at zero, eliminating any impossible negative sales forecasts.

To determine the optimal regularization parameter (α) for our Ridge regressions, we employed two distinct validation methodologies:

1. **Out-of-Sample (OOS) Validation:** We isolated the final 15 days of the training set to serve as a static validation holdout. The models were trained on all preceding data, and Bayes Search was utilized to select the α parameter that explicitly minimized the error (RMSLE) on this specific 15-day testing window.
2. **Time Series Cross-Validation (CV):** To prevent data leakage and evaluate performance across multiple time horizons, we used an expanding-window Time Series Split approach. We combined this with Bayes Search (and Grid Search for Model 3) to optimize α based on average performance across the folds. Crucially, the final 15 days of the training data were completely excluded from these CV folds.

After identifying the optimal α using both methods, we evaluated the models on the 15-day holdout set (Validation RMSLE). Finally, we retrained the models on the entirety of the training data and generated predictions for the Kaggle test set (Test RMSLE). The results are summarized in Table 1. Note that Model 1 (the baseline) has no tunable α or validation scores, so only its final test score is reported. Similarly, Model 3 lacks a single global α since it consists of many independent local models.

Table 1: Model Evaluation

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
OOS α	-	-	-	0.0023	559948.6743	0.0067	0.0010	223.2250
Validation RMSLE	-	0.4906	0.4600	0.5200	0.5389	0.5090	0.5094	0.5091
Test RMSLE	1.31791	0.72757	0.67004	0.46885	0.48015	0.46614	0.46792	0.47324
CV α	-	-	-	1952.1126	313963.0962	2111.7017	3332.6314	2728.3037
Validation RMSLE	-	-	0.4914	0.5369	0.5407	0.5352	0.5331	0.5301
Test RMSLE	-	-	0.62904	0.47383	0.47735	0.46953	0.46998	0.46822

As expected, the OOS Validation RMSLE is consistently lower than the CV Validation RMSLE, because the OOS α is explicitly tuned to minimize error on that exact 15-day validation window. In contrast, the CV approach never exposes the model to the validation set during hyperparameter tuning. Interestingly, with the exception of Model 5, the optimal OOS α is significantly smaller than the CV α , yet the final Test RMSLE scores remain comparable.

Ultimately, while Model 6 with the OOS α achieved the best Test RMSLE score (the primary metric of the competition), it is likely overfit to the specific temporal dynamics of the final 15 days of August. Based on the data in Table 1, Model 8 using the CV-derived α proved to be the most robust choice: it achieved a highly competitive Test RMSLE without explicitly overfitting to a single, static validation window.

5 Summary

To build the most robust and generalizable model for predicting store sales, we recommend utilizing a global Ridge regression approach (Model 8) with a cross-validated regularization parameter of $\alpha = 2728.3037$. This optimal model predicts `log1p(sales)` utilizing the following feature set:

1. Product family (categorical).
2. Store number (categorical).
3. Day of the week (categorical).
4. Sine and cosine transformations of the day of the year, utilizing periods of 1, 1/2, and 1/3 of a year to capture seasonality (numeric).
5. A boolean indicator for dates falling within one month of the April 16, 2016 earthquake (categorical).
6. The number of days since the last public sector payday (numeric).
7. Calendar holiday indicators (categorical).
8. 16-, 21-, and 28-day lags of `log1p(sales)` (numeric).
9. An interaction term between the product family and the number of items on promotion (categorical).
10. An interaction term between the product family and the store number (categorical).

During the inference phase, the model outputs predictions in the transformed logarithmic space. To finalize the forecasts, one must first clip the outputs to bound all predictions at zero (eliminating negative predictions), and subsequently apply the inverse log transformation (`expm1`) to project the values back to their original sales scale.